

VAYUSETU: Dataset Inventory

National Observation Data Sources & Integration Schema

1. Multi-Source Ingestion & Spatial Resolutions

VAYUSETU ingests real data spanning multiple spaceborne sensors, meteorological models, and ground station grids. The table below represents the current dataset inventory utilized by the twin state manager:

Dataset	Source Agency	Spatial Resolution	Temporal Resolution	Role in Twin State
IMD Rainfall Grid	IMD	0.25° × 0.25° (~25 km)	Daily	Primary precipitation training & validation
IMD Temperature Grid	IMD	1.0° × 1.0° (~100 km)	Daily	Max/Min temperature baselines
INSAT-3D LST	MOSDAC/ISRO	4 km (Thermal Infra)	Hourly	Land surface temperature assimilation
INSAT-3D SST	MOSDAC/ISRO	4 km (Thermal Infra)	Hourly	Sea surface temperature for ocean-land coupling
INSAT-3D Rainfall	MOSDAC/ISRO	4 km (Infrared)	Hourly	Real-time satellite rainfall estimates
Bhuvan LULC Grid	NRSC/ISRO	56 meters (1:250k)	Annual	Land Use classification & runoff coefficient matrix
NICES Soil Moisture	NICES/ISRO	0.25° × 0.25°	Daily	Soil moisture fraction and albedo calibration
GFS Forecasts	NCEP/NOAA	28 km Grid	3-Hourly	Numerical Weather Prediction baseline forecast

2. Data Ingestion & Live Connectors Pipeline

All raw datasets are downloaded via automated Cron/HTTP triggers, verified for checksums, and parsed using H5Py or NetCDF4 libraries. If satellite tiles are partially obscured by cloud cover, the Data Coverage Engine flags missing grid cells, and the twin state fills gaps using deterministic spatial interpolation (inverse distance weighting) before feeding the Kalman Filter.